Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -1(Adv-2020-P1-Model)** **Date: 26-04-2026**

Time: 09.00Am to 12.00Pm **GTA-27** **Max. Marks: 180**

KEY SHEET

PHYSICS

1	**B**	2	**C**	3	**D**	4	**B**	5	**A**	6	**B**
7	**ABCD**	8	**ACD**	9	**CD**	10	**BC**	11	**AC**	12	**ABC**
13	**3**	14	**2**	15	**6**	16	**4**	17	**2**	18	**4**

CHEMISTRY

19	**A**	20	**B**	21	**D**	22	**C**	23	**B**	24	**C**
25	**CD**	26	**ACD**	27	**ABC**	28	**ABCD**	29	**ABC**	30	**ABCD**
31	**6**	32	**3**	33	**104**	34	**31.38 to 31.40**	35	**297 to 303**	36	**5.70 to 5.90**

MATHEMATICS

37	**C**	38	**A**	39	**B**	40	**A**	41	**B**	42	**D**
43	**AC**	44	**ABD**	45	**AC**	46	**ACD**	47	**BCD**	48	**ABCD**
49	**1**	50	**0.7**	51	**1**	52	**1**	53	**32**	54	**0**

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SOLUTIONS

PHYSICS

1. On rising up as g decreases the time period of the pendulum increases and clock will slow down and start loosing time, lost per unit time can be given as dT/T = (1/2)dg/g here $g = GM / R^2$ and verify that option (B) is correct.

2. At $4^\circ C$ density of water is maximum so when heat is conducted and temperature of water at different regions become $4^\circ C$, it settles down at bottom and when temperature further falls below $4^\circ C$ its density decreases an it will float above so $0^\circ C$ will attain first at the region in between and at top.

3. $$\frac{1}{R_{total}} = \frac{1}{R_1 + R_2} + \frac{1}{R_3}$$

$$\frac{1}{R_{total}} = \frac{1}{\frac{L}{k_1 A} + \frac{L}{k_2 A}} + \frac{1}{\frac{2L}{k_3 \frac{A}{n}}}$$

$$= \frac{k_1 k_2 A}{L(k_1 + k_2)} + \frac{k_3 A}{2Ln}$$

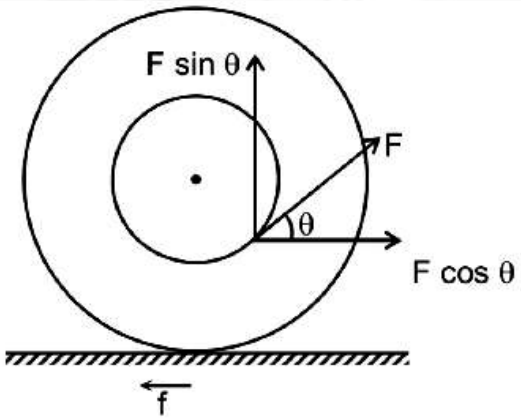
$$= \frac{2k_1 k_2 n + k_3(k_1 + k_2)}{2n(k_1 + k_2)} \frac{A}{L}$$

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4. Surface energy $\propto$ Area and rate of heat flow $\propto \frac{1}{Area}$

5. $F\cos\theta - f = ma_c$ ..... (i)

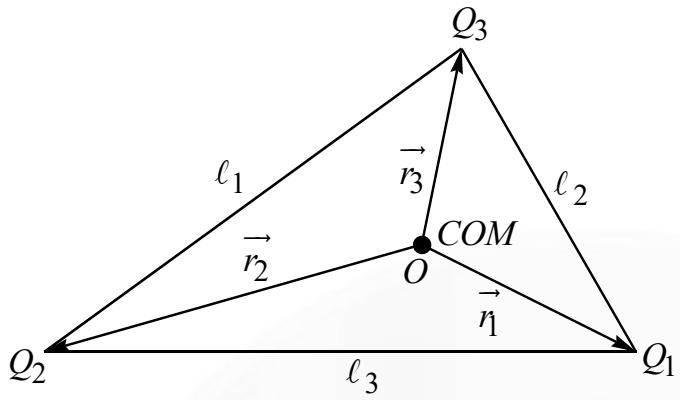
$f(2R) - FR = 2mR^2\alpha$ ..... (ii)



$a_c = \alpha 2R$

$\therefore f = \frac{F}{3}(1 + \cos\theta) \Rightarrow$ always leftward.

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6. Let COM to be origin

Net force Q_1 is $\vec{F_1} = \frac{KQ_1Q_2}{\ell_3^3}\left(\vec{r_1}-\vec{r_2}\right)+\frac{KQ_1Q_3}{\ell_2^3}\left(\vec{r_1}-\vec{r_3}\right)$

$$\vec{F_1} = KQ_1\left[\frac{Q_2}{\ell_3^3}+\frac{Q_3}{\ell_2^3}\right]\vec{r_1} - KQ_1\left[\frac{Q_2}{\ell_3^3}\vec{r_2}+\frac{Q_3}{\ell_2^3}\vec{r_3}\right] \quad \ldots\ldots\ldots (1)$$

$$m_1\vec{r_1}+m_2\vec{r_2}+m_3\vec{r_3}=\vec{O}\Rightarrow\vec{r_3}=-\frac{1}{m_3}\left(m_1\vec{r_1}+m_2\vec{r_2}\right) \quad \ldots\ldots\ldots (2)$$

Substituting (2) in (1),

$$\vec{F_1} = KQ_1\left[\frac{Q_2}{\ell_3^3}+\frac{Q_3}{\ell_2^3}\right]\vec{r_1} - \frac{KQ_1Q_2}{\ell_3^3}\vec{r_2}+\frac{KQ_1Q_3}{m_3\ell_2^3}\left(m_1\vec{r_1}+m_2\vec{r_2}\right)$$

$$= KQ_1\left[\frac{Q_2}{\ell_3^3}+\frac{Q_3}{\ell_2^3}\left(1+\frac{m_1}{m_3}\right)\right]\vec{r_1} + KQ_1\left[\frac{Q_3}{\ell_2^3}\cdot\frac{m_2}{m_3}-\frac{Q_2}{\ell_3^3}\right]\vec{r_2}$$

Charges move along fixed straight lines if $\vec{F_1}$ is along $\vec{r_1}$, $\vec{F_2}$ is along $\vec{r_2}$ and $\vec{F_3}$ is along $\vec{r_3}$. So $\vec{r_2}$ term should be zero in the above $\vec{F_1}$ equation. For that, $\frac{Q_3}{\ell_2^3}\cdot\frac{m_2}{m_3}=\frac{Q_2}{\ell_3^3}\Rightarrow\frac{Q_3}{m_3}\ell_3^3=\frac{Q_2}{m_2}\ell_2^3$

Similarly with $\vec{F_2}, \vec{F_3}$ expressions, we get a general conclusion as

$$\frac{Q_1}{m_1}\ell_1^3=\frac{Q_2}{m_2}\ell_2^3=\frac{Q_3}{m_3}\ell_3^3=\text{constant}$$

$$\Rightarrow \ell_1^3:\ell_2^3:\ell_3^3=\frac{m_1}{Q_1}:\frac{m_2}{Q_2}:\frac{m_3}{Q_3}$$

$$\ell_1:\ell_2:\ell_3=\sqrt[3]{\frac{m_1}{Q_1}}:\sqrt[3]{\frac{m_2}{Q_2}}:\sqrt[3]{\frac{m_3}{Q_3}}$$

$$=\sqrt[3]{27}:\sqrt[3]{8}:\sqrt[3]{64}$$

$$= 3:2:4$$

7. (i) At this time the two masses are travelling parallel to the x–axis in the centre of mass system and are both crossing the y–axis, one in each direction. This situation must be superimposed on that due to the

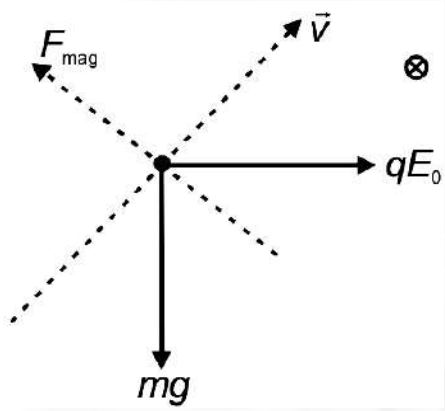
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motion of the centre of mass, which has moved along the line $x = L/2$ through a distance L and has a speed of V/2 in the y–direction. Block A is at $\left(\frac{1}{2}L, \frac{1}{2}L\right)$ with velocity $\left(\frac{1}{2}V, \frac{1}{2}V\right)$; block B is at $\left(\frac{1}{2}L, \frac{3}{2}L\right)$ with velocity $\left(-\frac{1}{2}V, \frac{1}{2}V\right)$.

(ii) Establish that the centre of mass motion is cyclic with period 8L/V. Block A is at (L, 50L) with velocity (0, V); block B is at (0, 50L) at rest.

8. No external torque is acting in this case too.
So, Kepler's law of areas is obeyed in this case too.

9. At time t_0, for velocity to be constant



$\vec{F}_{net} = 0$

$\therefore \vec{F}_{may} = -q\vec{E}_0 - m\vec{g}$

$\therefore \vec{v}$ should be at 45° with the horizontal

Also, $v_y = \frac{qE_0}{qB_0} = \frac{E_0}{B_0}$ and $v_x = \frac{mg}{qB_0} = \frac{E_0}{B_0}$

Also, $v_x = \frac{E_0}{B_0} = \frac{qE_0t_0}{m} = gt_0$ and $v_y = v - gt_0$

$\frac{E_0}{B_0} = v_0 - gt_0$

$gt_0 = v_0 - gt_0$

$\therefore v_0 = 2gt_0 = \frac{2E_0}{B_0}$

10. $\alpha + \theta = 90^\circ$

$t_1 = \frac{2u\cos\theta}{g}, t_2 = \frac{2u\cos\theta}{g}, \frac{2u^2\sin\theta\cos\theta}{g} = 60m$

$t_1t_2 = \frac{4u^2\sin\theta\cos\theta}{g^2} = 4 \times \frac{30}{10} = 12s$

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$(t_1+t_2)^2=(t_1-t_2)^2+4t_1t_2=1+48$

$t_1+t_2=7s$

$\therefore t_1=4s$ and $t_2=3s$

$\tan\theta=\frac{4}{3}\simeq 53^\circ$

$\therefore \alpha=37^\circ$ and $\theta-\alpha=16^\circ$

Also, $u=25m/s$

11. $x_{cm}=\frac{\int x\lambda dx}{\int dm}=\frac{\int_0^L x\lambda_0\left(1+\frac{x}{L}\right)dx}{\int_0^L \lambda_0\left(1+\frac{x}{L}\right)dx}=\frac{5L}{9}$

$m=\int dm=\frac{3\lambda_0 L}{2}$

$\left[\left(A\cdot\frac{L}{9}\right)\rho g\right]\frac{17}{18}L\sin\theta=F\cdot L\cos\theta$

12. $\frac{Q_{out}}{Q_{in}}=\frac{T_{out}}{T_{in}}$ and $Q_{out}=Q_{in}+W$

$\Rightarrow Q_{in}=W\left(\frac{T_L}{T_H-T_L}\right)$

$\frac{Q_{in}}{t}=P\left(\frac{T_L}{T_H-T_L}\right)$

13. Since $\frac{\mu}{4}<\frac{\mu}{2}<\mu$, the upper block will move faster than the middle block and hence force of friction on upper block is towards left.





$\therefore f=\frac{\mu}{4}mg \ \therefore a_A=\frac{\mu g}{4}$ left ward

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$$f' = \frac{\mu}{2}\left(m + \frac{m}{2}\right)g = \frac{3}{4}\mu mg$$

$\therefore a_B = \mu g$ (Leftward) $\qquad \therefore a_{A/B} = \frac{3}{4}\mu g$ (Rightward)

For sliding down the block B, A has to move a distance $\frac{3L}{8}$ relative to block B.

$$\frac{3L}{8} = \frac{1}{2}\frac{3}{4}\mu g t^2 \qquad \therefore \qquad t = \sqrt{\frac{L}{\mu g}} = 3\text{sec}$$

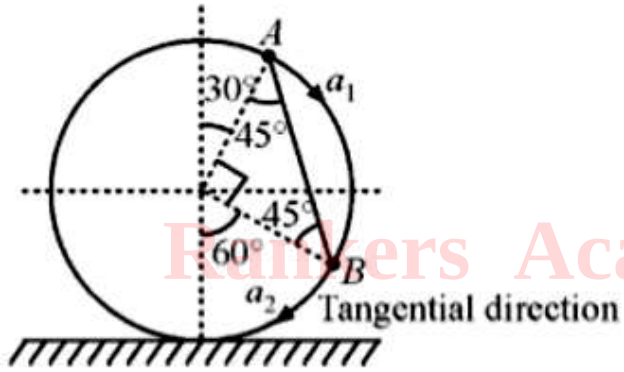
14. Let a_1 and a_2 be the tangential accelerations of two beads respectively. Then

For bead A

$$m_1 g \sin 30^\circ + T \sin 45^\circ = m_1 a_1 \qquad \text{..... (i)}$$

For bead B

$$m_2 g \sin 60^\circ - T \sin 45^\circ = m_2 a_2 \qquad \text{..... (ii)}$$



Since string remains taut $\qquad \therefore a_1 \cos 45^\circ = a_2 \cos 45^\circ$

(Acceleration along string remains same)

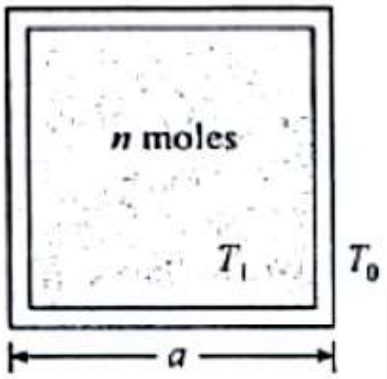
$\Rightarrow a_1 = a_2 = a \qquad \therefore$ From (i) and (ii)

$$a = \frac{g\left(m_1 + \sqrt{3}m_2\right)}{2\left(m_1 + m_2\right)}; \frac{T}{\sqrt{2}} = \frac{m_1 g\left(m_1 + \sqrt{3}m_2\right)}{2\left(m_1 + m_2\right)} - \frac{m_1 g}{2}$$

$$T = \frac{\left(\sqrt{3} - 1\right)\left(m_1 m_2 g\right)}{\sqrt{2}\left(m_1 + m_2\right)}$$

15. As shown in figure if at time $t = 0$, gas temperature is T_1 and after time t, its temperature is T, heat will be conducted through the walls of container, the rate of heat conducted to outside surrounding can be written as

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$$\frac{dQ}{dt}=\frac{k\left(6a^2\right)}{x}\left[T-T_0\right] \qquad \text{…….. (i)}$$

If dQ heat is conducted outside in a time dt and due to this if the temperature of gas falls by dT, we have

$dQ=-nC_v dT$ [As volume of container is constant]

or $dQ=-\frac{nR}{\gamma-1}dT \left[As\ C_V=\frac{R}{\gamma-1}\right]$ …….. (ii)

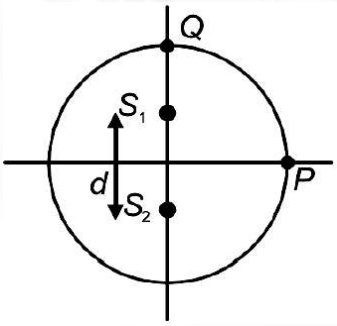
From equation (i) and (ii) we have

$$\frac{nR}{\gamma-1}\frac{dT}{dt}=-\frac{6ka^2}{x}\left(T-T_0\right) \text{ or } \frac{dT}{T-T_0}=\frac{6ka^2\left(\gamma-1\right)}{nRx}dt$$

Integration the above expression within proper limits from beginning $\left(t=0\right)$, we get

$$T=T_0+\left(T_1-T_0\right)e^{\frac{6ka^2\left(\gamma-1\right)}{nRx}t}$$

16. The wavelength of sound wave in air is



$$\lambda=\frac{v}{f}=\frac{320}{16\times10^3}=2\times10^{-2}m$$

For point P,

Path different $\Delta x_P=0$

For point Q,

$\Delta x_Q=2m=n\lambda$

$\therefore n=100$

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Which means 400 maximas will be heard by the person riding in the train in time

$$t = \frac{400}{2} = 200\,s$$

$\therefore$ Length of the circular track = $v \times t$

$= 20 \times 200$

= 4000 m

= 4 km

17. Using Biot–Savart law,

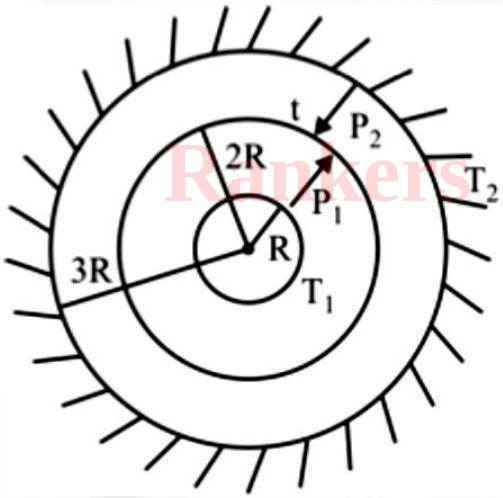
$$\vec{B} = \frac{\mu_0 I(\sin\alpha + \sin\beta)}{4\pi r}$$

$$B_P = \frac{\mu_0 I}{4\pi d} + \frac{\mu_0 I(\sin\alpha + \cos\alpha)}{4\pi d\cos\alpha} + \frac{\mu_0 I(\tan\alpha)}{4\pi d} = \frac{\mu_0 I}{4\pi d}(1+\tan\alpha)2$$

and $B_Q = \frac{\mu_0 I(\sin\alpha + \cos\alpha)}{4\pi d\cos\alpha} = \frac{\mu_0 I(1+\tan\alpha)}{4\pi d}$

$$\frac{B_P}{B_Q} = 2$$

18.



$$P_1 = \sigma\left(4\pi R^2\right)\left(T_1^4\right)$$

$$P_2 = \sigma\left(4\pi\left(4R^2\right)\right)\left(T_2^4\right)$$

$$P_{emission} = \sigma 4\pi R^2\left(t^4\right) + \sigma 4\pi\left(4R^2\right)t^4$$

$$T_1^4 + 4T_2^4 = 5t^4$$

$$t = \left(\frac{T_1^4}{5} + \frac{T_2^4}{\frac{5}{4}}\right)^{\frac{1}{4}}$$

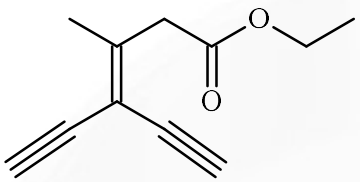
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$$x=5, y=\frac{5}{4} \Rightarrow \frac{x}{y}=4$$

CHEMISTRY

19. Cu^{+} do not undergo oxidation in N_2 atm.

20. In case of CN^{-} LUMO is π antibonding and in case of O_2 LUMO is σ antibonding.



21. ethyl 4-ethynyl-3-methylhex-3-en-5-ynoate

22. Compound IV (terminal alkyne) is acidic due to the high % s-character (sp) of carbon stabilising the conjugate base $\rightarrow$ A is correct.
Compound II (cyclopentadiene) is more acidic than I (triphenylmethane) because its conjugate base is aromatic (6π electrons) $\rightarrow$ B is correct.
Correct acidity order is: V (Phenol) > II (cyclopentadiene) > IV (alkyne) > I (triphenylmethane) > III (alkane) Given order in C places water incorrectly relative to IV, hence C is incorrect. Electron-withdrawing NO_2 group increases acidity of III via −I effect $\rightarrow$ D is correct.

23. $3\times E^{\circ}=1\times 0.96+2\times 1.04 \Rightarrow E^{\circ}=1.01V$

24. I represents dissociative adsorption with no energy of activation.
II represents dissociative adsorption with energy of activation.

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25. $H_3B^{-1}-C\overleftarrow{=}O^{+1}$

26. Flux is used to remove impurities.

27. D-Threose is an **aldotetrose** with **two chiral centres (C-2 and C-3)**.
Relations are decided by comparing **configurations at both chiral centres**.
A: Both chiral centres have the **same configuration** as D-threose $\rightarrow$ **Identical. Correct**
B: Both chiral centres are **inverted** relative to D-threose $\rightarrow$ **Enantiomer. Correct**
C: Only **one chiral centre is inverted**, the other is same $\rightarrow$ **Diastereomer. Correct**
D: Only one centre is inverted, so it is **not an enantiomer** $\rightarrow$ **Incorrect**

28. All four Newman projections P, Q, R and S represent the same molecule, viewed along the same C–C bond but drawn in different staggered conformations and rotations.

29. Mean free path of a gas is given by: $\lambda=\frac{kT}{\sqrt{2}\pi d^2 P} \Rightarrow \lambda \propto \frac{T}{d^2 P}$

A) At same T and P, smaller molecular diameter d gives larger λ. Correct.

B) λ depends on number density n, not on molecular speed. Correct.

C) When $\lambda >> d$, intermolecular interactions are negligible and gas behaves ideally. Correct.

D) At constant volume, $P \propto T$. Hence $\lambda \propto \frac{T}{P}$ =constant, not inversely proportional to T. Incorrect.

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30. A) For the ground state (1s), the wavefunction is maximum at the nucleus; hence the probability density is maximum at $r = 0$. Correct.

B) For the ground state, the radial probability distribution has only one peak at $r = \frac{a_0}{z}$. Correct.

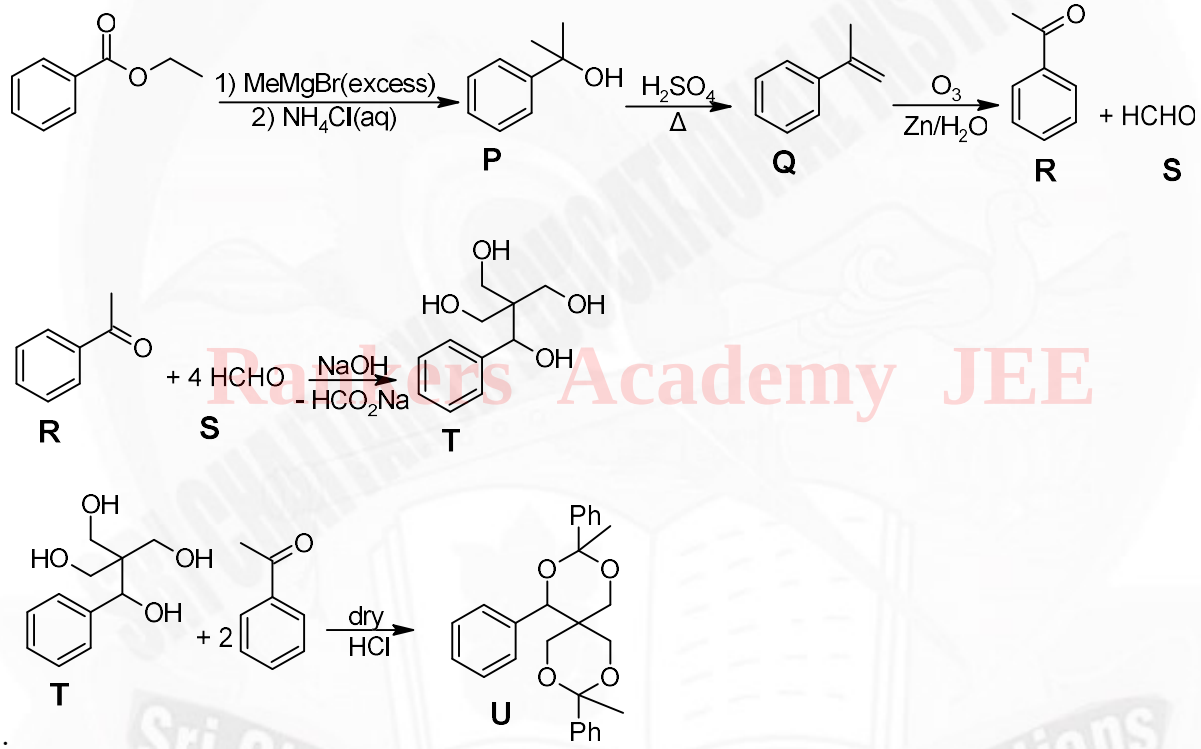
C) The number of radial nodes depends on n and $\ell\left(= n - \ell - 1\right)$ and is independent of nuclear charge Z. Correct.

D) For the ground state, the position of maximum radial probability varies as $\frac{1}{z}$; hence when $Z \to 2Z$, it becomes half. Correct.

31. $Vo^{+2}, I_2^-, Cu^{+2}, O_3^-, O_2$ and S_2 are paramagnetic.

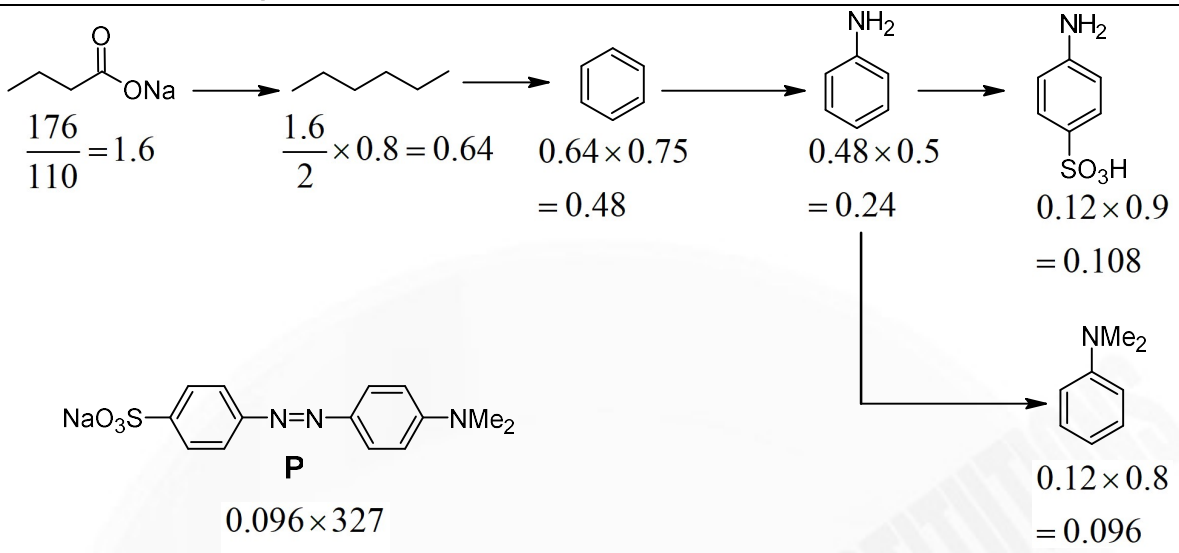
32. $\left[Ni\left(NH_3\right)_6\right]^{+2}, \left[Co\left(H_2O\right)_6\right]^{+2}, \left[Cu\left(H_2O\right)_6\right]^{+2}$

33.



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34.

35. For $NaC\ell$ (rock-salt structure), number of formula units per unit cell: $Z = 4$

Due to Schottky defects, only 90% units remain: $Z_{eff} = 4 \times 0.9 = 3.6$

Out of 100 units, 98 are $NaC\ell$ and 2 are $MgC\ell_2$.

$$M_{eff} = \frac{2}{100} \times 94 + \frac{98}{100} \times 58 = 58.72$$

$$d = \frac{Z \times M_{eff}}{N_A a^3} \Rightarrow 1.63 = \frac{3.6 \times 58.72}{6 \times 10^{23} \times a^3} \Rightarrow a^3 = 216 \times 10^{-24}\,cm^3 \Rightarrow a = 600pm$$

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For $NaC\ell$, shortest distance: $d = \frac{a}{2} = 300\,pm$

$$\Delta S^\circ = nF\frac{\partial E}{\partial T} = 6 \times 96500 \times \frac{0.01}{400} = 14.475J$$

$$\Delta G^\circ_1 = \Delta H^\circ - T_1 \Delta S^\circ$$

36. $$-6 \times 96500 \times 0.02 = \Delta H^\circ - 400 \times 14.475$$

$$\Rightarrow \Delta H^\circ = -6 \times 96500 \times 0.02 + 400 \times 14.475 = -5790 Jmol^{-1}$$

MATHEMATICS

37. $s^2 = at^2 + 2bt + c$

Differentiating w.r.t. t

$$2s\frac{ds}{dt} = 2at + 2b$$

$$s^2 \cdot \left(\frac{ds}{dt}\right)^2 = (at + b)^2 = a^2t^2 + 2abt + b^2$$

$$s^2 \cdot \left(\frac{ds}{dt}\right)^2 = as^2 + b^2 - ac; \qquad \left(\frac{ds}{dt}\right)^2 = a + \frac{b^2 - ac}{s^2}$$

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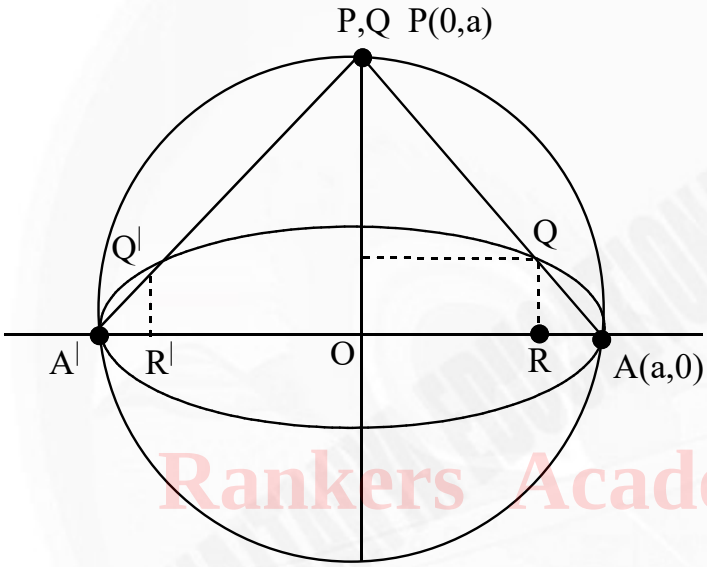
Differentiating w.r.t t again

$$2\frac{ds}{dt}\cdot\frac{d^2s}{dt^2}=0-2\frac{b^2-ac}{s^3}\frac{ds}{dt}\Rightarrow\frac{d^2s}{dt^2}=\frac{ac-b^2}{s^3}$$

38. $$\frac{AP}{AQ}+\frac{A'P}{A'Q'}=3$$

$$\frac{AQ+PQ}{AQ}+\frac{A'Q'+PQ'}{A'Q'}=3$$

$$\frac{PQ}{AQ}+\frac{PQ'}{A'Q'}=1$$



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Both same ratios (by sym.)

$$\frac{PQ}{AQ}=\frac{1}{2}\Rightarrow Q\left(\frac{a}{3},\frac{2a}{3}\right)$$

As Q satisfy ellipse equation, we get $\frac{a^2}{b^2}=2$

$$\Rightarrow e=\frac{1}{\sqrt{2}}$$

39. $$a=\min\sum_{r=0}^{4}674\left(\left|z-\frac{e^{\frac{ir\pi}{2}}}{100}\right|+\left|zw-\frac{e^{\frac{ir\pi}{2}}}{1011}\right|+\left|zw^2-\frac{e^{\frac{ir\pi}{2}}}{1011}\right|\right).$$

As we know, $|x-y|\geq\big||x|-|y|\big|$

$\Rightarrow$ min. will occur at $z=0$

$$a_{\min}=674\times\frac{3}{1011}\times5=10$$

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40. Consider the expansion of $\left(\frac{2}{3}-\frac{1}{3}\right)\left(\frac{4}{5}-\frac{1}{5}\right)\left(\frac{6}{7}-\frac{1}{7}\right).....\left(\frac{2n}{2n+1}-\frac{1}{2n+1}\right)$ the negative terms corresponds to an odd number of tails. So product is (Probability of even – probability of odd) The product reduces to $\frac{1}{2n+1}$ obviously probability even + probability odd = 1.

$\Rightarrow$ Probability odd = $\frac{1-\frac{1}{2n+1}}{2}=\frac{n}{2n+1}$

41. $f(1)+f(2)+f(3)+f(4)=2000$

$A=f(1)+2f(2)+3f(3)+4f(4)$

$\frac{f(1)+2f(2)+3f(3)+4f(4)}{4}\geq\left[24f(1)\,f(2)\,f(3)\,f(4)\right]^{1/4}$

$A^4\geq 256\times 24\,f(1)\,f(2)\,f(3)\,f(4)$

$\Rightarrow A^4\geq 6144\,f(1)\,f(2)\,f(3)\,f(4)$

$\therefore f(1)=2f(2)=3f(3)=4f(4)=k$

$\Rightarrow f(2)=\frac{k}{2},f(3)=\frac{k}{3},f(4)=\frac{k}{4}$

$\Rightarrow k+\frac{k}{2}+\frac{k}{3}+\frac{k}{4}=2000\Rightarrow k=960$

$f(1)=960,f(2)=480,f(3)=320,f(4)=240$

$g(x)=k(x-\alpha)(x-\beta)(x-\gamma)(x-\delta),k=1$

$g(i)=k(i-\alpha)(i-\beta)(i-\gamma)(i-\delta),k=1$

$\Rightarrow|g(i)|=\sqrt{\left(1+\alpha^2\right)\left(1+\beta^2\right)\left(1+\gamma^2\right)\left(1+\delta^2\right)}$

Given $|g(i)=1\Rightarrow\alpha=\beta=\gamma=\delta=0|$

$\therefore g(x)=x^4$

$\Rightarrow f(1)=d,f(2)=c,f(3)=b,f(4)=a$

$\Rightarrow d=960,c=480,b=320,a=240$

42. (A) $g(x)=f(f(x))$

$g(a)=f(f(a))=f(b)=a$

$g(b)=f(f(b))=f(a)=b$

$g'(k)=\frac{g(b)-g(a)}{b-a}=\frac{b-a}{b-a}=1,k\in(a,b)$

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Clearly $f'(f(k)).f'(k)=1$

Case–1: If $f(k)\neq k$ consider $k=c$ $f(k)=d$

Given statement is true $f'(c)\cdot f'(d)=1$

Case–2: If $f(k)=k$ consider two interval $(a,k),(k,b)$

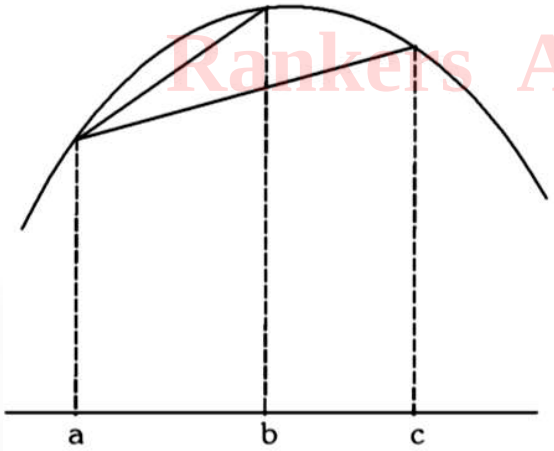
Apply LMVT $f'(c)\cdot f'(d)=\dfrac{f(k)-f(a)}{k-a}\cdot\dfrac{f(b)-f(k)}{b-k}=1$

(B) Apply LMVT in two intervals $\left(a,\dfrac{a+b}{2}\right),\left(\dfrac{a+b}{2},b\right)$

$$\frac{f\left(\frac{a+b}{2}\right)-f(a)}{\left(\frac{a+b}{2}\right)-a}=f'(c_1)\Rightarrow\frac{f\left(\frac{a+b}{2}\right)-f(a)}{\left(\frac{b-a}{2}\right)}=f'(c_1)$$

$$2\left(\frac{f(b)-f(a)}{b-1}\right)=f'(c_1)+f'(c_2)$$

(C) $\dfrac{f(b)-f(a)}{b-a}>\dfrac{f(c)-f(a)}{c-a}$



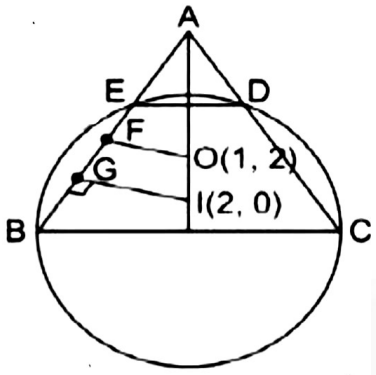
(D) $\lim\limits_{x\to0^+}\left(\dfrac{x^{x^{x^x}}-x^{x^x}}{x^x-x}\right)=\dfrac{1-0}{1-0}=1$

43. Since A lies on line IO

$\Rightarrow A(-4,12)$

$AI=6\sqrt{5},AO=5\sqrt{5},$

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Let G and F are mid points of BE and BA.

$$\Rightarrow AB = 2AF = 2\cdot AO\cdot\cos 30^\circ = 2\cdot 5\cdot\sqrt{5}\cdot\frac{\sqrt{3}}{2} = 5\cdot\sqrt{15}$$

$$AG = AI\cos 30^\circ = 3\cdot\sqrt{15}$$

Now, $AG = \dfrac{AE+AB}{2} \Rightarrow AE = 2AG - AB$

$$= 2\cdot 3\cdot\sqrt{15} = 5\cdot\sqrt{15} = \sqrt{15}$$

$$\Delta = \frac{\sqrt{3}}{4}\cdot\left(AB^2 - AE^2\right) = 90\cdot\sqrt{3}$$

44. $\displaystyle\int_0^1 \left(f(x)\right)^2 dx + \int_0^1 \left(x^2 - x\right)h'(x)dx$

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$$= \int_0^1 \left(f(x)\right)^2 dx + \left[\left(x^2 - x\right)h(x)\right]_0^1 - \int_0^1 (2x-1)\left(h(x)\right)dx$$

$$= \int_0^1 \left(f(x)\right)^2 dx - \int_0^1 (2x-1)\left(f(x) - 9x + 4\right)dx$$

After solving the above expression, we get

$$= 1 + \int_0^1 \left(f(x)\right)^2 dx \left(\because \int_0^1 \left(f(x)\right)dx > 0\right)$$

45. Let's say there are n + 10 teams

Bottom 10 teams won 45 games/points against each other

Top n teams won $\dfrac{n(n-1)}{2}$ points against each other

There are 10n games/points exchanged between the top n teams and the bottom 10 teams.

So, $45 + \dfrac{n(n-1)}{2} = 10 \Rightarrow n = 25$

46. Conics are $x^2 - xy = 1$ and $x^2 - xy = -1$ hence they are conjugate hyperbolas of each other.

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$$\Rightarrow \frac{1}{e_1^2}+\frac{1}{e_2^2}=1$$

Asymptotes are $x=0$ and $x=y$ lines

$$\Rightarrow \left(\frac{b}{a}\right)_2=\tan\frac{\pi}{8}=\sqrt{2}-1 \text{ and } \left(\frac{b}{a}\right)_1=\tan\frac{3\pi}{8}=\sqrt{2}+1$$

47. (A) $\lim\limits_{x\to 0}\dfrac{x-\sin x-\cos^{-1}\left(e^{\frac{-x^4}{2}}\right)}{x^2}=\lim\limits_{x\to 0}\dfrac{x-\sin x-\sin^{-1}\sqrt{1-e^{-x^4}}}{x^2}=-1$

(B) $\lim\limits_{x\to 0}\dfrac{\cos^{-1}\left(e^{\frac{-x^4}{2}}\right)-x\tan^{-1}\left(\frac{2x}{1+x^2}\right)}{x^2}=-1$

(C) $\lim\limits_{x\to 0}\dfrac{2x-\sin 2x-\tan^{-1}\left(\frac{2x^3}{1+x^6}\right)}{x^3}=\frac{4}{3}-2=\frac{-2}{3}$

(D) $\lim\limits_{x\to 0}\dfrac{\frac{\pi}{2}-\left(\tan^{-1}\left(\tan^{-1}\left(\frac{2x}{1+x^2}\right)\right)\right)-2\cos^{-1}\left(e^{\frac{-x^4}{2}}\right)}{\cos^{-1}\left(\frac{1-\cos\frac{1}{x}}{\frac{1}{x}}\right)}=\dfrac{\frac{\pi}{2}-2\frac{\pi}{2}}{\frac{\pi}{2}}=-1$

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48. (A) $\dfrac{1}{2}\sum\limits_{n=1}^{10}\left(\dfrac{3^n}{5^n-3^n}-\dfrac{3^{n+1}}{5^{n+1}-3^{n+1}}\right)=\dfrac{1}{2}\left[\dfrac{3}{2}-\dfrac{3^{11}}{5^{11}-3^{11}}\right]$

(B) $\dfrac{\left(\underbrace{31^x+31^x....}_{y\ times}\right)+\left(\underbrace{65^y+65^y....}_{x\ times}\right)}{x+y}\geq\left(31^{xy}65^{xy}\right)^{\frac{1}{x+y}}$

$$\frac{y\cdot 31^x+x\cdot 65^y}{x+y}\geq 31\times 65 \qquad \left(As\ \frac{x+y}{xy}=1\right)$$

$$\frac{31^x}{x}+\frac{65^y}{y}\geq 2015$$

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(C) $t_r=(r+1)^2, \sum_{r=1}^{\infty}\frac{1}{r(r+1)}=1$

(D) $(c-0)^2=4(a-b)(b-c)$

$(c+a)^2=4(a+c)-4b^2$

$(a+c-2b)^2=0$

49. $y^3y''=-1$

$y'y''=\frac{-y'}{y^3}$

Integrating both sides

$\frac{(y')^2}{2}=\frac{1}{2y^2}+C$

Putting $x=1, y=1$ and $y'(1)=0$

$0=\frac{1}{2}+C\Rightarrow C=\frac{-1}{2}$

$\frac{(y')^2}{2}=\frac{1}{2y^2}-\frac{1}{2}$

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$(y')^2=\frac{1}{y^2}-1\Rightarrow(y')^2=\frac{1-y^2}{y^2}\Rightarrow y'=\pm\frac{\sqrt{1-y^2}}{y}\Rightarrow\frac{y}{\sqrt{1-y^2}}y'=\pm1$

Taking as +ve sign

$\int\frac{y}{1-y^2}y'dx=\int 1dx$

Putting $y=t\Rightarrow y'dx=dt$

$\int\frac{t}{\sqrt{1-t^2}}dt=x+\lambda\Rightarrow 1-t^2=z^2\Rightarrow -2tdt=2zdz$

$-\int\frac{zdz}{z}=x+\lambda\Rightarrow -z=x+\lambda\Rightarrow-\sqrt{1-t^2}=x+\lambda\Rightarrow-\sqrt{1-y^2}=x+\lambda$

Putting $x=1, y(1)=1$

$\therefore\lambda=-1\Rightarrow\sqrt{1-y^2}=x-1\Rightarrow 1-y^2=x^2-2x+1$

$y^2=2x-x^2$

$y=\sqrt{2x-x^2}=f(x)$ $\quad\{\because y\geq 0\forall x\in D_f\}$

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50. $\int_0^1 \left(f^2\left(x^2\right)-2x\cdot f\left(x^2\right)+x^2\right)\cdot dx=0$

$\Rightarrow f\left(x^2\right)=x$

$\Rightarrow f\left(x\right)=\sqrt{x}$

51. $G=\left[{}^nC_0\cdot{}^nC_1\cdot{}^nC_2\cdot{}^nC_3.....\right]^{1/n}\le\frac{{}^nC_0+{}^nC_1+{}^nC_2+....{}^nC_n}{n}=\frac{2^n}{n}$

$\Rightarrow G^{1/n}\le\frac{2}{n^{1/n}}$

$\Rightarrow \lim_{n\to\infty} G^{1/n}<\lim_{n\to\infty}\frac{2}{n^{1/n}}=\frac{2}{1^+}=2^-$

(As $\lim_{n\to\infty} n^{1/n}=1$)

52. Let $Lz=\frac{\cos\theta+i\sin\theta}{\cos\theta};z^k=\frac{\cos k\theta+i\sin k\theta}{\cos^k\theta}$

$$\sum_{k=0}^{n-1} z^k=\frac{1-z^n}{1-z}=\frac{1-\frac{\cos n\theta+i\sin n\theta}{\cos^n\theta}}{1-\frac{\cos\theta+i\sin\theta}{\cos\theta}}=\frac{\sin n\theta}{\sin\theta\cos^{n-1}\theta}+i\frac{\cos^n\theta-\cos n\theta}{\sin\theta\cos^{n-1}\theta}$$

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So, $\text{Re}\left(\sum_{k=0}^{n-1} z^k\right)=\frac{\sin n\theta}{\sin\theta\cos^{n-1}\theta}=\frac{\sin n\frac{\pi}{6}}{\sin\frac{\pi}{6}\left(\cos\frac{\pi}{6}\right)^{n-1}}$

53. $\sum_{K=1}^{64} K\alpha-\sum_{K=1}^{64}\{K\alpha\}$ is divisible by 64

$\Rightarrow \alpha\cdot\frac{64\cdot 65}{2}-\sum_{K=1}^{64}\{K\alpha\}$

Here, $0\le\sum_{K=1}^{64}\{K\alpha\}<64$ is never divisible by 64.

$\Rightarrow \alpha$ has to be even integer.

Also, $\left(a^2+2b\right)^2=65-8b^2$

$=\{57, 33\}$ hence no solution.

54. $P_n=\frac{5}{6}\cdot P_{n-1}+\frac{1}{6}\cdot\frac{5}{6}\cdot P_{n-2}+\frac{1}{6}\cdot\frac{1}{6}\cdot\frac{5}{6}\cdot P_{n-3}$

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